

Phase 9 CVRP Suite Report

Overview

- Condition count: 6.
- Best held-out condition: phase9_closeout_baseline_clarke_wright_savings.

Condition Summary

Condition	Mode	Held-out Feasibility	Held-out Penalized Gap	Held-out Feasible Gap	Held-out Runtime (ms)	Accepted Epochs	Fallback Epochs	Generation Errors	Mean Novelty	Mean Complexity
phase9_closeout_baseline_nearest_neighbor_constructive	baseline_nearest_neighbor_constructive	1.0	0.273435	0.273435	42.55776	0	0	0	0.0	0.0
phase9_closeout_baseline_clarke_wright_savings	baseline_clarke_wright_savings	1.0	0.073766	0.073766	76.12722	0	0	0	0.0	0.0
phase9_closeout_baseline_regret_insertion_local_search	baseline_regret_insertion_local_search	1.0	0.466391	0.466391	1360.19192	0	0	0	0.0	0.0
phase9_closeout_direct_generate_plus_one_repair	direct_generate_plus_one_repair	0.0	13.06	n/a	135.26838	1	0	0	0.968411	15.15
phase9_closeout_budget_matched_no_replay	budget_matched_no_replay	0.0	3.0	n/a	204.11066	2	0	0	0.940383	15.08
phase9_closeout_replay_solver_evolution	replay_solver_evolution	1.0	0.273435	0.273435	141.26554	0	0	0	0.0	0.0

Condition Notes

phase9_closeout_baseline_nearest_neighbor_constructive

- Execution mode: baseline_nearest_neighbor_constructive.
- Train feasibility rate: 1.0.
- Train penalized gap: 0.337704.
- Held-out feasibility rate: 1.0.
- Held-out penalized gap: 0.273435.
- Held-out runtime ms: 42.55776.
- Accepted epoch count: 0.
- Fallback epoch count: 0.

- Generation-error epoch count: 0.
- Robustness dispersion across held-out structure families: 0.002777.
- Worst held-out instances:
- X-n247-k50: feasible=True, penalized gap=0.408032, errors=[]
- X-n176-k26: feasible=True, penalized gap=0.364218, errors=[]
- X-n223-k34: feasible=True, penalized gap=0.317111, errors=[]
- X-n190-k8: feasible=True, penalized gap=0.145642, errors=[]
- X-n275-k28: feasible=True, penalized gap=0.132172, errors=[]

phase9_closeout_baseline_clarke_wright_savings

- Execution mode: baseline_clarke_wright_savings.
- Train feasibility rate: 1.0.
- Train penalized gap: 0.064783.
- Held-out feasibility rate: 1.0.
- Held-out penalized gap: 0.073766.
- Held-out runtime ms: 76.12722.
- Accepted epoch count: 0.
- Fallback epoch count: 0.
- Generation-error epoch count: 0.
- Robustness dispersion across held-out structure families: 0.007791.
- Worst held-out instances:
- X-n247-k50: feasible=True, penalized gap=0.108521, errors=[]
- X-n176-k26: feasible=True, penalized gap=0.099285, errors=[]
- X-n190-k8: feasible=True, penalized gap=0.061249, errors=[]
- X-n275-k28: feasible=True, penalized gap=0.057708, errors=[]
- X-n223-k34: feasible=True, penalized gap=0.042065, errors=[]

phase9_closeout_baseline_regret_insertion_local_search

- Execution mode: baseline_regret_insertion_local_search.
- Train feasibility rate: 1.0.
- Train penalized gap: 0.451049.
- Held-out feasibility rate: 1.0.
- Held-out penalized gap: 0.466391.
- Held-out runtime ms: 1360.19192.
- Accepted epoch count: 0.
- Fallback epoch count: 0.
- Generation-error epoch count: 0.
- Robustness dispersion across held-out structure families: 0.086899.
- Worst held-out instances:

- X-n275-k28: feasible=True, penalized gap=0.746105, errors=[]
- X-n223-k34: feasible=True, penalized gap=0.650864, errors=[]
- X-n247-k50: feasible=True, penalized gap=0.395235, errors=[]
- X-n176-k26: feasible=True, penalized gap=0.390341, errors=[]
- X-n190-k8: feasible=True, penalized gap=0.149411, errors=[]

phase9_closeout_direct_generate_plus_one_repair

- Execution mode: direct_generate_plus_one_repair.
- Train feasibility rate: 0.0.
- Train penalized gap: 8.958333.
- Held-out feasibility rate: 0.0.
- Held-out penalized gap: 13.06.
- Held-out runtime ms: 135.26838.
- Accepted epoch count: 1.
- Fallback epoch count: 0.
- Generation-error epoch count: 0.
- Robustness dispersion across held-out structure families: 1.616667.
- Worst held-out instances:
- X-n275-k28: feasible=False, penalized gap=15.7, errors=["NameError: name 'isinstance' is not defined", 'missing 274 customers']
- X-n247-k50: feasible=False, penalized gap=14.3, errors=["NameError: name 'isinstance' is not defined", 'missing 246 customers']
- X-n223-k34: feasible=False, penalized gap=13.1, errors=["NameError: name 'isinstance' is not defined", 'missing 222 customers']
- X-n190-k8: feasible=False, penalized gap=11.45, errors=["NameError: name 'isinstance' is not defined", 'missing 189 customers']
- X-n176-k26: feasible=False, penalized gap=10.75, errors=["NameError: name 'isinstance' is not defined", 'missing 175 customers']

phase9_closeout_budget_matched_no_replay

- Execution mode: budget_matched_no_replay.
- Train feasibility rate: 0.0.
- Train penalized gap: 3.0.
- Held-out feasibility rate: 0.0.
- Held-out penalized gap: 3.0.
- Held-out runtime ms: 204.11066.
- Accepted epoch count: 2.
- Fallback epoch count: 0.
- Generation-error epoch count: 0.
- Robustness dispersion across held-out structure families: 0.0.

- Worst held-out instances:
- X-n176-k26: feasible=False, penalized gap=3.0, errors=['solver did not return a list of routes']
- X-n190-k8: feasible=False, penalized gap=3.0, errors=['solver did not return a list of routes']
- X-n223-k34: feasible=False, penalized gap=3.0, errors=['solver did not return a list of routes']
- X-n247-k50: feasible=False, penalized gap=3.0, errors=['solver did not return a list of routes']
- X-n275-k28: feasible=False, penalized gap=3.0, errors=['solver did not return a list of routes']

phase9_closeout_replay_solver_evolution

- Execution mode: replay_solver_evolution.
- Train feasibility rate: 1.0.
- Train penalized gap: 0.337704.
- Held-out feasibility rate: 1.0.
- Held-out penalized gap: 0.273435.
- Held-out runtime ms: 141.26554.
- Accepted epoch count: 0.
- Fallback epoch count: 0.
- Generation-error epoch count: 0.
- Robustness dispersion across held-out structure families: 0.002777.
- Worst held-out instances:
- X-n247-k50: feasible=True, penalized gap=0.408032, errors=[]
- X-n176-k26: feasible=True, penalized gap=0.364218, errors=[]
- X-n223-k34: feasible=True, penalized gap=0.317111, errors=[]
- X-n190-k8: feasible=True, penalized gap=0.145642, errors=[]
- X-n275-k28: feasible=True, penalized gap=0.132172, errors=[]

Judge Note

Phase-9 CVRP Closeout Suite Summary (Conservative)

Condition	Feasibility Rate (Heldout)	Heldout Feasible Gap	Runtime (ms)	Robustness Dispersion	Notes
phase9_closeout_baseline_nearest_neighbor_constructive	100%	0.2734	42.6	0.0028	Baseline, fast runtime, moderate gap and robustness
phase9_closeout_baseline_clarke_wright_savings	100%	**0.0738 (Best)**	76.1	0.0078	Best heldout gap, baseline, slightly longer runtime
phase9_closeout_baseline_regret_insertion_local_search	100%	0.4664	1360.2	0.0869	Slowest, worst gap, baseline
phase9_closeout_direct_generate_plus_one_repair	0%	0.0	135.3	1.6167	Zero feasibility, large penalized gap (13.06), high complexity and novelty
phase9_closeout_budget_matched_no_replay	0%	0.0	204.1	0.0	Zero feasibility, penalized gap 3.0, learned condition, no replay

Condition	Feasibility Rate (Heldout)	Heldout Feasible Gap	Runtime (ms)	Robustness Dispersion	Notes
phase9_closeout_replay_solver_evolution	100%	0.2734	141.3	0.0028	Feasibility matched baseline nearest neighbor, no gap improvement over learned or baseline, replay condition

Key Points:

- **Baselines performed well** with perfect feasibility and the Clarke-Wright savings baseline achieved the best heldout objective gap (0.0738).
- **Learned conditions did not clearly beat the fixed baselines**:
- The direct synthesis (direct_generate_plus_one_repair) failed feasibility entirely and had a large objective gap.
- Budget-matched independent search without replay also had zero feasibility, disqualifying it from outperforming baselines.
- **Replay-aware iterative search condition** (replay_solver_evolution) achieved full feasibility and similar gap as nearest neighbor baseline but did **not improve upon baseline or learned no-replay control** in objective gap.
- Replay was **better in feasibility and robustness than budget-matched no replay**, which had zero feasibility, indicating replay improves feasibility but does not yet reduce the objective gap below baselines or no-replay learned conditions under the current setup.

Conclusions

- The **best heldout condition by objective gap** was the Clarke-Wright savings baseline.
- No learned or replay condition clearly outperformed the best fixed baseline.
- **Replay improved feasibility over budget-matched no-replay learned control**, but no gap improvements were observed.
- Learned methods showed higher complexity and novelty but failed to produce feasible solutions or better gaps within the evaluation timeframe.

Overall conservative assessment: Baseline heuristics currently dominate in the phase-9 CVRP closeout suite by solution quality and feasibility. Replay aids feasibility compared to learned no-replay, but learned methods still lag behind fixed baseline performance in objective gap and robustness.